DC Circuit Part 2

Circuit Connections

Continuation...

Series Circuit

a) The same current flows through all the components like resistances.

$$I_T = I_1 = I_2 = I_3$$

b) The total resistance equals the sum of the resistances connected in series.

$$R_T = R_1 + R_2 + R_3$$

c) There will be voltage drop across each component like a resistance given by Ohm's law and the sum of the voltage drops is equal to the applied voltage.

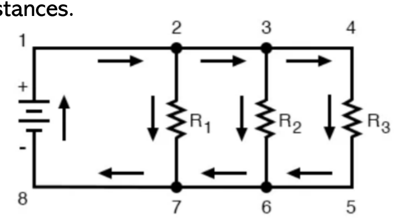
$$E = V_1 + V_2 + V_3$$

d) The total power equals the sum of the individual power of each resistance.

$$P_T = P_1 + P_2 + P_3$$

Parallel Circuit

Parallel circuit is a circuit in which one end of each resistance is joined to a common point and the other end of each resistance is joined to another common point so that there are as many paths for current flow as the number of resistances.



a) Parallel voltages are equal.

$$E = V_1 = V_2 = V_3 = ... = V_n$$

b) The total current is equal to the sum of the branch currents.

$$I_T = I_1 + I_2 + I_3 + ... + I_n$$

c) The reciprocal of the total resistance is equal to the sum of the reciprocals of the resistances connected in parallel.

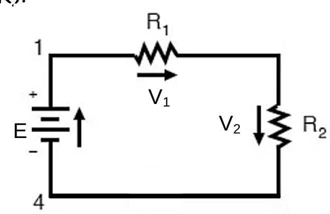
$$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + ... + \frac{1}{R_n}$$

d) The total power is the sum of the individual power of the components in the parallel circuit (similar to in series circuit).

$$P_T = P_1 + P_2 + P_3 + ... + P_n$$

Voltage Division Theorem (VDT)

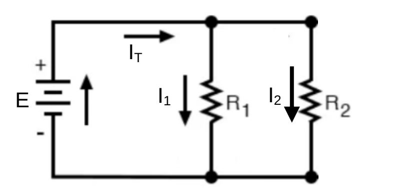
This theorem is applicable to series circuits where the total source voltage is given and voltages across the resistances are required. Provided that component values are given, the voltage across each can be directly determined without even knowing the current flowing. This theorem is also known as the **voltage divider rule** (VDR).



$$V_1 = \frac{R_1}{R_1 + R_2} E \qquad V_2 = \frac{R_2}{R_1 + R_2} E$$

Current Division Theorem (CDT)

This theorem is intended for parallel circuits where the current flowing in each branch is required. The total current and component values must be given to directly find the individual branch current. This theorem is also called the **current divider rule** (CDR).



$$I_1 = \frac{R_2}{R_1 + R_2} I_T \qquad I_2 = \frac{R_1}{R_1 + R_2} I_T$$

Important terms in Network Laws & Theorems

Network is the interconnection of components such as resistors and batteries forming a complicated circuit.

Branch (b) represents a single element such as a voltage source or a resistor

Node (n) is the point of connection between two or more branches.

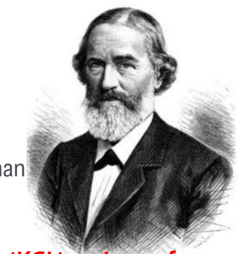
Loop (ℓ**)** is any closed path in a circuit.

Mesh is a loop which does not contain any other loops within it.

The fundamental theorem of network topology is given by

$$b = \ell + n - 1$$

Kirchhoff's Laws



These laws were first described in 1845 by German physicist Gustav Kirchhoff.

Kirchhoff's Current Law (KCL) or Law of Conservation of Current

The first law states that:

> "In any electrical network, the algebraic sum of the currents meeting at a point (or junction) is zero."

This law was based from the law of conservation of charge. The law can also be stated as:

> "incoming currents to a junction or point = outgoing currents from that junction or point".

To make the sum of entering and leaving currents zero the sign convention followed is that **entering or incoming current is positive and leaving or outgoing current is negative**.

Kirchhoff's Voltage Law (KVL) or the Law of Conservation of Voltage

According the second law:

> "The algebraic sum of the products of currents and resistances in each of the conductors in any closed path (or mesh) in a network plus the algebraic sum of the emfs in that path is zero."

The second law was based from the law of conservation of energy. The statement can be mathematically expressed by the following equation:

$$\Sigma IR + \Sigma emf = 0 \text{ round a mesh}$$

Nodal Method

This method offers the advantage of requiring minimum number of equations needed to be written to determine desired quantities. This method is using the current law alone.

> "Every junction in the network that represents a connection of three or more branches is regarded as a node. One node is always considered a reference node or zero-potential point, then current equations are written for the remaining junctions using Kirchhoff's current-law; thus; a solution is possible with n-1 equations, where n is the number of nodes."